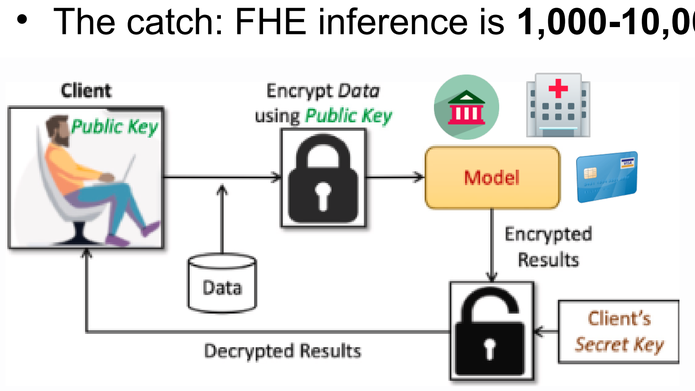
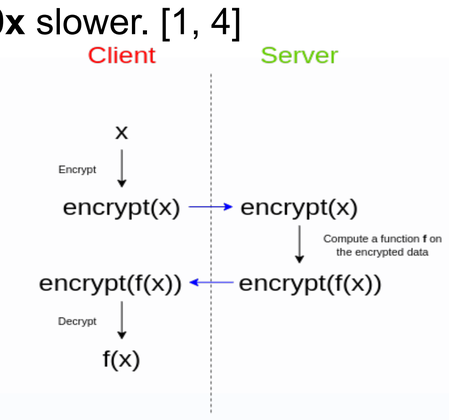
Joint Bit-Width and Fan-In Sensitivity Optimization for FHE-Aware Neural Network Acceleration

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INTRODUCTION

- Fully Homomorphic Encryption (FHE) lets the cloud compute on encrypted data, so the data is never exposed.
- It unlocks AI on sensitive data: encrypted medical diagnosis, private fraud and credit scoring, biometric matching, private search.
- The catch: FHE inference is **1,000-10,000x** slower. [1, 4]





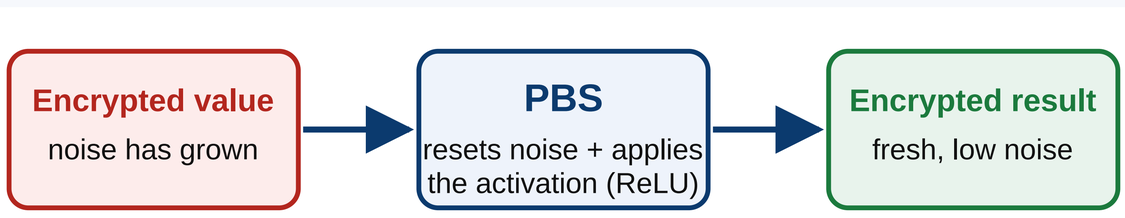
MOTIVATION

Where the time goes (**TFHE Bottleneck**): Every neuron's activation needs one bootstrapping (PBS) call [1], and each call gets exponentially costlier as the integer bit-width grows. Speed is therefore controlled by two knobs: precision(bits) and pruning(neurons).

The Accumulator Trap: Accumulator width expands with incoming fan-in. In standard TFHE compilers (e.g., Zama Concrete ML [3]), this exponential scaling forces baselines into crippling uniform 2-bit precision (W2A2) to avoid circuit explosion.

Unstructured Pruning Fails: Prior TFHE compression pipelines [4] rely on heuristic unstructured pruning, which sets individual weights to zero but leaves physical neuron counts,

Onerous Manual Tuning: Finding the right layer-wise precision and sparsity currently requires manual trial-and-error



CONTRIBUTIONS

1. **Automated optimizer**

Encrypted AI no longer needs an FHE expert and weeks of trial-and-error compiling. A hospital or bank team can deploy a private model by stating a speed target.

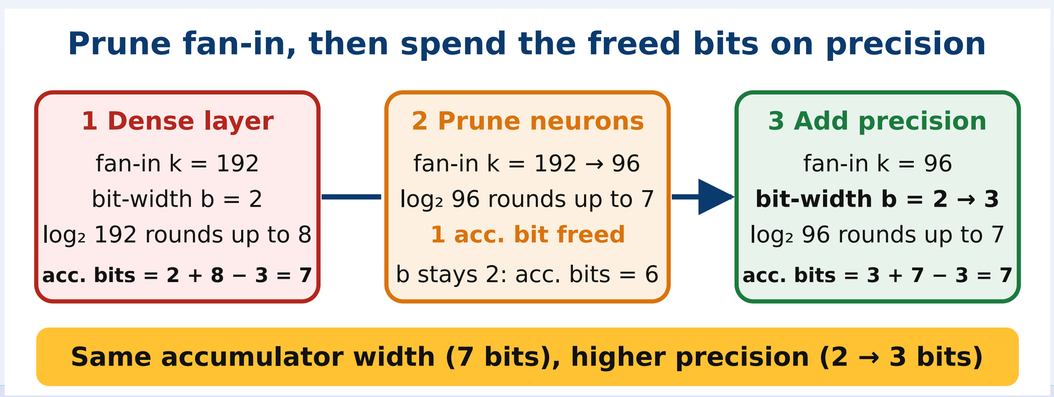
2. **Structured pruning**

Each removed neuron = one fewer PBS call. [1]

Bootstrapping is the main cost of encrypted inference, so fewer calls means less compute per private prediction. That is the path to making private diagnosis, fraud scoring or biometric matching affordable at scale

3. **Spend the savings**

Pruning shrinks fan-in, freeing bits to spend on precision. Privacy usually costs accuracy. Here encrypted models get cheaper and more accurate, which matters most where errors are costly



APPROACH

Our main goal is to minimize total distortion such that:

Total PBS cost ≤ budget B

$$\min_{\{b_l,s_l\}} \sum_{l=1}^{L} D_l(b_l, s_l) \quad \text{s.t.} \quad \sum_{l=1}^{L} C_l \le B$$

$$b_l \in \{3,4,5,6,8\}, s_l \in \{0, 0.25, 0.5, 0.75\}$$

PBS Operational Cost Calculation:

Bootstrapping cost doubles per accumulator bit

$$C_l = N_l(1 - s_l) \cdot 2^{b_{\text{acc}}(l)}, \quad b_{\text{acc}}(l) \approx b_l + \lceil \log_2 k_l \rceil$$ [3]

with $k_l = \lceil N_{\text{in},l}(1 - s_{l-1}) \rceil$

Hessian Distortion Calculation:

Combines quantization noise and pruning loss

$$D_l = Tr(H_l) \cdot 2^{-2b_l} + \sum_{i \in \text{pruned}} \left(\sum_j \frac{1}{2} H_{jj} w_{ij}^2 \right)$$

Computed matrix-free using Hutchinson trace estimation. [2]

Knapsack Dynamic Programming:

$$V(l, c) = \min_{b_l, s_l} \{ D_l + V(l + 1, c - C_l) \}$$

where $V(l,c)$ is the minimum distortion from layer l under remaining budget c. State transitions evaluate discrete b_l, s_l choices and account for inter-layer fan-in coupling s_{l-1} to yield the exact global Pareto frontier

RESULTS

- **4x Fewer Bootstrappings:** Structured pruning shrinks hidden layers (192 to 48), cutting expensive PBS operations from 384 to 96
- **+2.6% Higher Accuracy:** Reclaimed accumulator headroom unlocks 3-8 bits precision (vs. uniform 2-bit), reaching 94.8% vs. 92.2% at smaller accumulator size than 6 bits.

PBS calls (bar) · precision (fc1, fc2, fc3)	Latency / inference	Test accuracy (MNIST)
b_acc ≤ 6 bits (Low-Latency)		
384 PBS Zama (2,2,2)	31.0 s	92.2%
96 PBS Ours (3,3,4) · 4x fewer	7.7 s	94.8% (+2.6 pts)
b_acc ≤ 8 bits (High-Accuracy)		
384 PBS Zama (2,2,2)	300.0 s	97.1%
96 PBS Ours (4,8,6) · 4x fewer	75.0 s	98.2% (+1.1 pts)

Modeled FHE cost: up to 4.01x lower (2.37x–4.01x) at target accuracy; model fit on 42 Concrete ML circuits ($R^2 > 0.98$).
Single PBS: ≈ 80 ms at b_acc ≤ 6 · ≈ 780 ms at b_acc ≤ 8 (latency = per-image, single MNIST sample)
Setup: Intel Core i7-11800H @ 2.30 GHz (8 cores / 16 threads), 32 GB RAM; Concrete ML v1.9.0, Brevitas v0.10.2

CONCLUSION & IMPACT

- **Cloud Practicality:** Makes private inference with TFHE viable for regulated healthcare and fintech under interactive SLAs.
- **Broad Generality:** Extends directly to CKKS/BGV noise reduction, zk-ML, and FHE ASICs.

REFERENCES

Please see the paper via the QR Code for the references, which are numbered accordingly in this poster, as well as for details of notations, other experiments and further explanation.

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